

Jonah Saitz

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EDUCATION

Brown University – Sc.B. Design Engineering, 3.9/4.0 GPA | Providence, RI | **Expected Graduation May 2026**

Relevant Courses: Control Systems Engineering, Dynamics and Vibrations, Linear System Analysis, Thermodynamics

ENGINEERING EXPERIENCE

Nextera Robotics – Deployment Specialist | Pittsburgh, PA | November 2024 – Present

Autonomous mobile robots for construction site progress and hazard monitoring.

- Performed mechanical adjustments and hardware upgrades to improve robot stability and field performance.
- Diagnosed and resolved network connectivity issues affecting data transmission and image capture reliability.
- Assisted with sensor calibration and troubleshooting of LIDAR-based navigation and mapping subsystems during live deployments and demonstrations.
- Increased sitewide image capture rate from 71% to 94% (company average: 74%) through systematic field testing and hardware/network fixes.

Wilhelmus Lab, Brown University – Undergraduate Researcher | Providence, RI | September 2025 – Present

Bio-inspired autonomous underwater vehicles (AUVs) for propulsion research, turbulent navigation and control, and swarming.

- Tested nitinol wire antennas to sense turbulence-induced strain for improved underwater navigation and control.
- Built strain measurement system using strain gauges, Wheatstone bridge, and load cell amplifier to capture voltage fluctuations corresponding to antenna deflection.
- Developed Arduino sketches to collect sensor data, visualize, and validate laboratory tests.
- Designed and fabricated waterproof antenna and sensor housings in Fusion360, using both FDM and resin 3D printing to prototype, test, and iterate.

UViva, Brown University – Mechanical / Systems Lead | Providence, RI | September 2025 – December 2025

Low-cost wearable device to monitor UV exposure and relative hydration for outdoor workers.

- Measured microcontroller, sensors, wires, and battery to determine minimum enclosure dimensions.
- Designed enclosure in SolidWorks, using FDM 3D printing for fabrication.
- Iterated through 4 prototypes, adding features such as metal contacts for hydration sensing and a USB-C charging port.
- Integrated UV and hydration sensors into enclosure, positioning components to minimize electrical interference, ensure consistent skin contact, and provide continuous sensing.
- Built and tested breadboard prototypes with ESP32, UV, and hydration sensors, evaluating sensor response under blacklight, sunlight, and hydration experiments over multiple trials.
- Balanced affordability, comfort, and manufacturability, producing a functional prototype suitable for early-stage implementation in outdoor workplaces.

Space Engineering, Brown University – Avionics Team | Providence, RI | September 2022 – May 2025

Experimental CubeSat to test performance of perovskite solar cells in orbit.

- Designed satellite power and radio schematics in KiCAD to ensure reliable data collection and transmission.
- Assembled, wired, and soldered electronics, verifying system functionality with multimeter and oscilloscope testing.
- Tested power, control, and communication between CubeSat hardware and ground station as part of Avionics Team
- Conducted environmental testing of electronics and mechanical assemblies to evaluate durability under vibration, temperature, and mechanical stress.
- Assisted with management of satellite BOM and PCB ordering after validation of schematic designs.

LEADERSHIP & OUTREACH

Remix Lab, Environmental Charter School – Makerspace Assistant | Pittsburgh, PA | June 2025 – August 2025

- Authored SOPs and standardized makerspace procedures for 3D printers and laser cutters.
- Configured equipment software and managed makerspace inventory using spreadsheets.
- Designed and fabricated custom components in Fusion360 to improve equipment usability and safety.

Venture Outdoors – Outdoor Educator | Pittsburgh, PA | June 2023 – August 2024

- Led STEM-based outdoor lessons, including physics experiments and hands-on engineering activities.
- Managed large groups on kayaking, biking, and fishing excursions, building skills in leadership, public speaking, teamwork, and risk management.

SKILLS & INTERESTS

Technical Skills: CAD (SolidWorks, Fusion, Onshape), 3D printing (FDM, resin), MATLAB, Simulink, Python, PID control, microcontrollers (Arduino, ESP32, STM32), lab instrumentation (oscilloscope, multimeter, signal generator), KiCAD, Git/GitHub.

Interests: Robotics, space exploration, biotech, renewables, personal finance, weightlifting, outdoor recreation.